MACHINE LEARNING LAB · FINAL PROJECT PRESENTATION

A Multimodal Deep Learning Framework for Explainable, Multi-Horizon Load-Shedding Risk Early Warning in Bangladesh

Calibrated Low / Medium / High risk alerts for the next 1 and 3 days, for the national grid and 9 zones

Low | Medium | High | Uncertain

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INTRODUCTION

Load shedding is rising, yet planners only see the past

21,075

mmcf domestic gas output in FY2024, down from 24,993 in FY2020

25%

of the gas supply was imported LNG in FY2025, up from 7.3% in FY2020

43%+

of electricity still comes from gas-fired plants that are hit by fuel shortages

2.4% → 27%

High-risk unit-days in our data: 87 of 3,630 in 2025 vs 812 of 2,980 in 2026

Why existing work is not enough

- Bangladeshi studies forecast demand (a number in MW) — a regression task
- BPDB publishes a daily report, but it describes yesterday
- Planners need: how risky are the next days, why, and how sure is the model?

Who is affected

- Homes, factories, hospitals, schools
- Fuel scheduling and maintenance planning
- Climate: heat-driven demand peaks (SDG 7, SDG 13)

Source: Khatun et al., The Financial Express (2025); our dataset (BPDB reports).

PROBLEM STATEMENT

Our task: predict the risk of load shedding, not the demand

Input

the last 14 days of grid, weather and context data

10 units

national grid + 9 BPDB zones

Output

risk class at T+1 and T+3

Low Med High

How a day is labelled (for each unit u and day t)

$$\text{shedding ratio } r = \text{load shed (MW)} / \text{demand (MW)}$$

Low	no load shedding on that day
Medium	$0 < r \le \theta$ (θ = 7.6% for zones, 1.9% for the national grid)
High	$r > \theta$ θ = median of the non-zero ratios in the training years only

OBJECTIVE

Objectives and contributions

1

Risk warning, not demand forecasting

One multi-task network: Low / Medium / High at T+1 and T+3 for 10 units

2

Multimodal fusion with modality attention

Grid, weather and context encoders; ablations show which source matters

3

Trustworthy, explainable output

Temperature-scaled probabilities, an 'Uncertain' option, SHAP + main driver

4

Extended, reusable dataset

Public 2019–2024 data extended to 28 Oct 2026 with our own BPDB PDF parser

5

Honest evaluation + lab experiments

Strict 2025–26 test, PR-AUC, fusion & task ablations, walk-forward validation

6

Working system

Streamlit dashboard (live alert, time machine, what-if) + Kaggle notebook

LITERATURE REVIEW

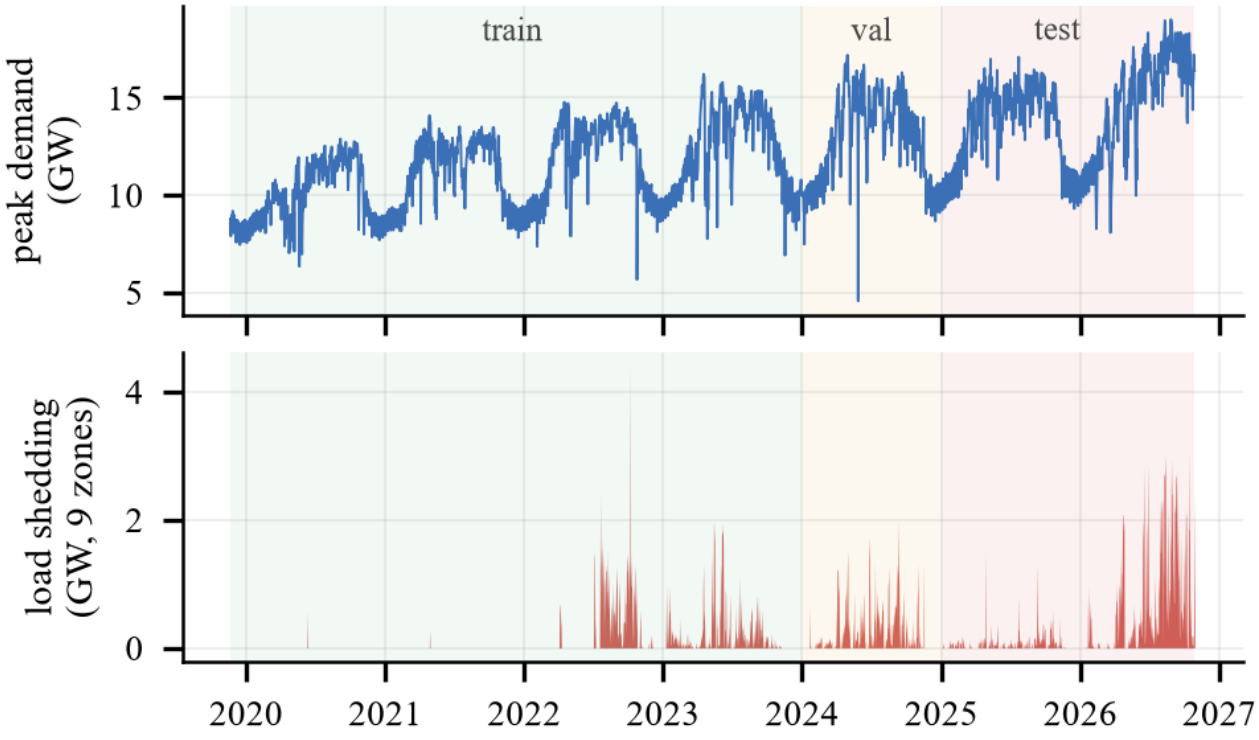
Literature review: prior work forecasts demand, not risk

Work	Task	Method / data	Limitation for our goal
Roy et al. (2021)	demand forecast	LSTM, hourly smart-grid data, 1.22% MAPE	regression, no risk class
Akhter et al. (2024)	Dhaka demand	CNN + stacked BiLSTM, 1.64% MAPE	one city, no warning
Islam et al. (2025)	national demand	LSTM, Prophet, RF + weather/holidays	demand only, no zones
Hossain et al. (2025)	dataset	daily BPDB data 2019–2024 (we extend it)	ends Dec 2024
Lim et al. TFT (2021)	multi-horizon	attention, interpretable	not for risk classes
Guo / Hendrickx	trust	temperature scaling, reject option	not applied to the grid
This project	**risk early warning**	**multimodal GRU + attention, calibrated, abstains**	**—**

Research gap: no study predicts a zone-level risk class at several horizons, fuses grid + weather + context, reports calibration and abstains, and tests on unseen divisions and a later period.

DATASET

Dataset: public data extended to October 2026



Top: national peak demand. Bottom: total load shedding of the 9 zones. Shaded: train / validation / test.

2,515 daily records (21 Nov 2019 – 28 Oct 2026)

10 units: national grid + 9 BPDB zones

24,920 leakage-safe 14-day samples

- Mendeley V4 (2019–2024): demand, supply, shedding, fuel limits
- BPDB / NLDC PDF reports (2024–2026): own scraper + parser
- NASA POWER weather for 9 zones · calendar (holidays, Ramadan, Eid)

DATASET

Preprocessing, quality control and leakage protection

1

Parse & date-check

printed date vs file name; parser matches V4 on 99.3–100% of zone values

2

Repair & merge

PDF value wins over V4 on overlap; demand rebuilt as evening-peak gen. + shedding (0.08% error)

3

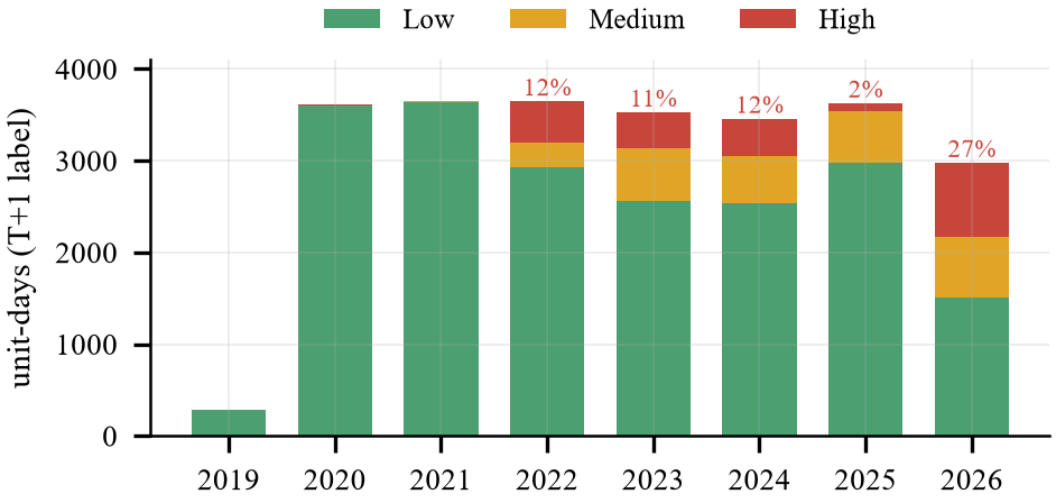
No future leakage

forward-fill ≤ 3 days, labels never filled, scaling and θ from training rows only

4

Automatic tests

scramble all test-period data → train/val features, labels, thresholds must not change

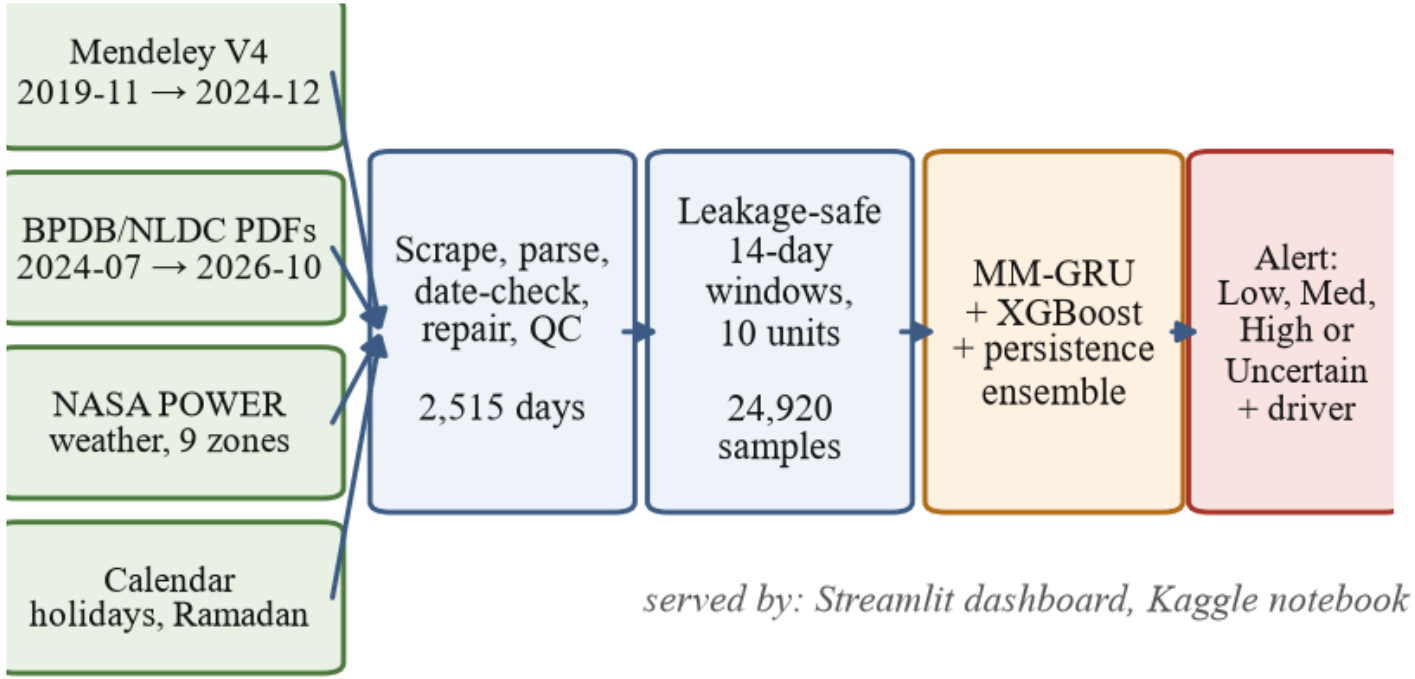


Chronological split (by the T+3 target date)

Train: Dec 2019 – Dec 2023 · 14,770 samples · 5.8% High
Validation: 2024 · 3,480 samples · 12.0% High
Test: 2025 – Oct 2026 · 6,660 samples · 13.5% High

METHODOLOGY

Methodology: end-to-end workflow



Same inputs and the same chronological split for every model.

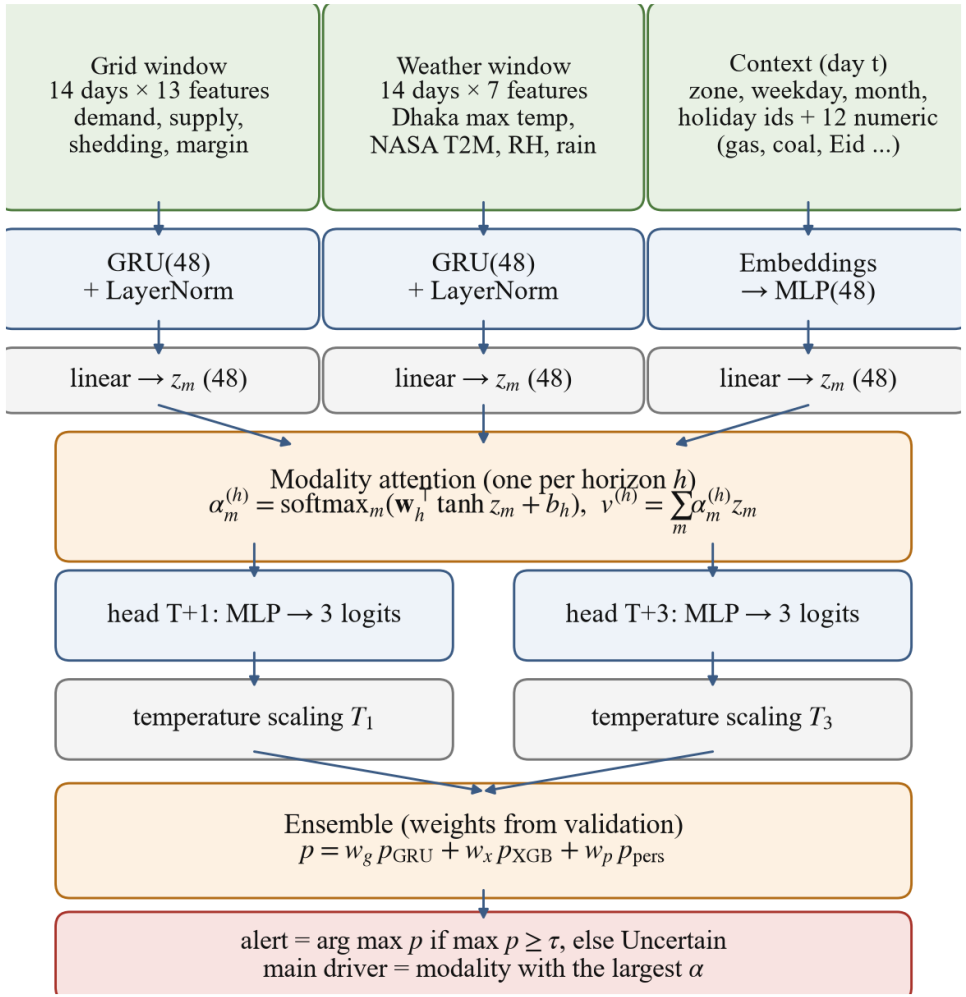
Every choice (early stopping, temperature, τ, ensemble weights) is made on validation; the test period is used once.

Models compared

- Persistence: tomorrow = today (hard to beat!)
- BPDB rule: depth-2 tree on reserve margin
- Logistic regression, Random forest
- XGBoost on 107 flattened features
- Multimodal GRU (ours), 3 seeds
- Ensemble: GRU + XGBoost + persistence, weights chosen on validation

METHODOLOGY

Multimodal GRU with modality attention



Three encoders

GRU(48) for the 14×13 grid window, GRU(48) for the 14×7 weather window, embeddings + MLP for context

Modality attention

one attention layer per horizon weighs grid / weather / context; the largest weight = main driver

Two heads, one network

multi-task: T+1 and T+3 heads share the encoders; only 31,780 parameters

Training

class-weighted cross-entropy (High days are rare), AdamW, early stopping on validation macro-F1, 3 seeds

METHODOLOGY

Trust layer: calibration, abstention and the ensemble

1

Temperature scaling

$p = \mathrm{softmax}(z / T)$

one T per horizon, fitted on validation → confidence matches accuracy

2

Abstain option

if $\max p < \tau \rightarrow$ “Uncertain”

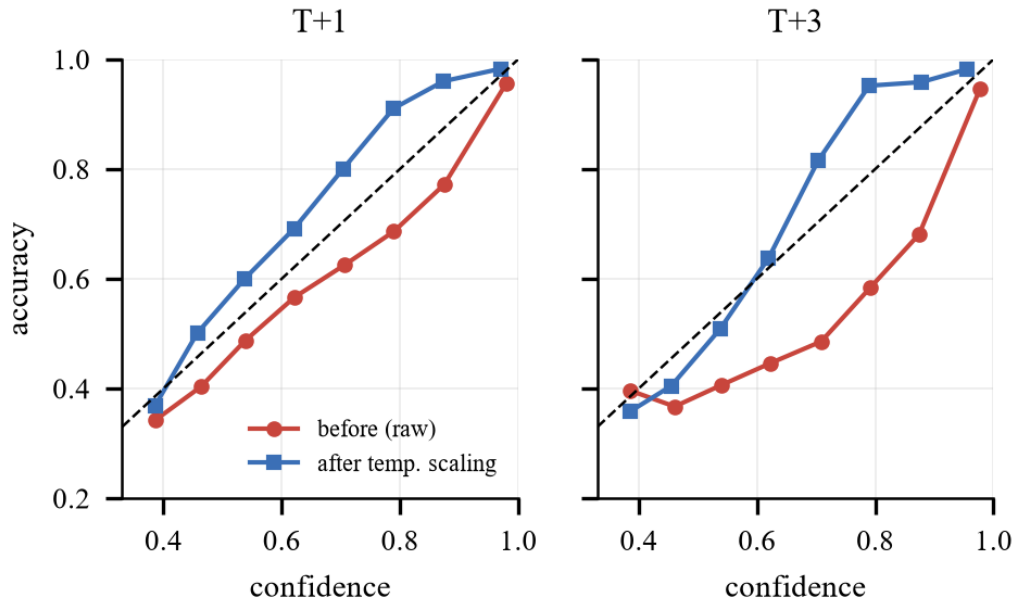
τ chosen on validation (answer ≥ 70%), shows the best guess

3

Ensemble

$p = w_g \cdot GRU + w_x \cdot XGB + w_p \cdot persistence$

weights on validation: T+1 0.1/0.6/0.3, T+3 0.3/0.5/0.2



ECE (calibration error) of the GRU

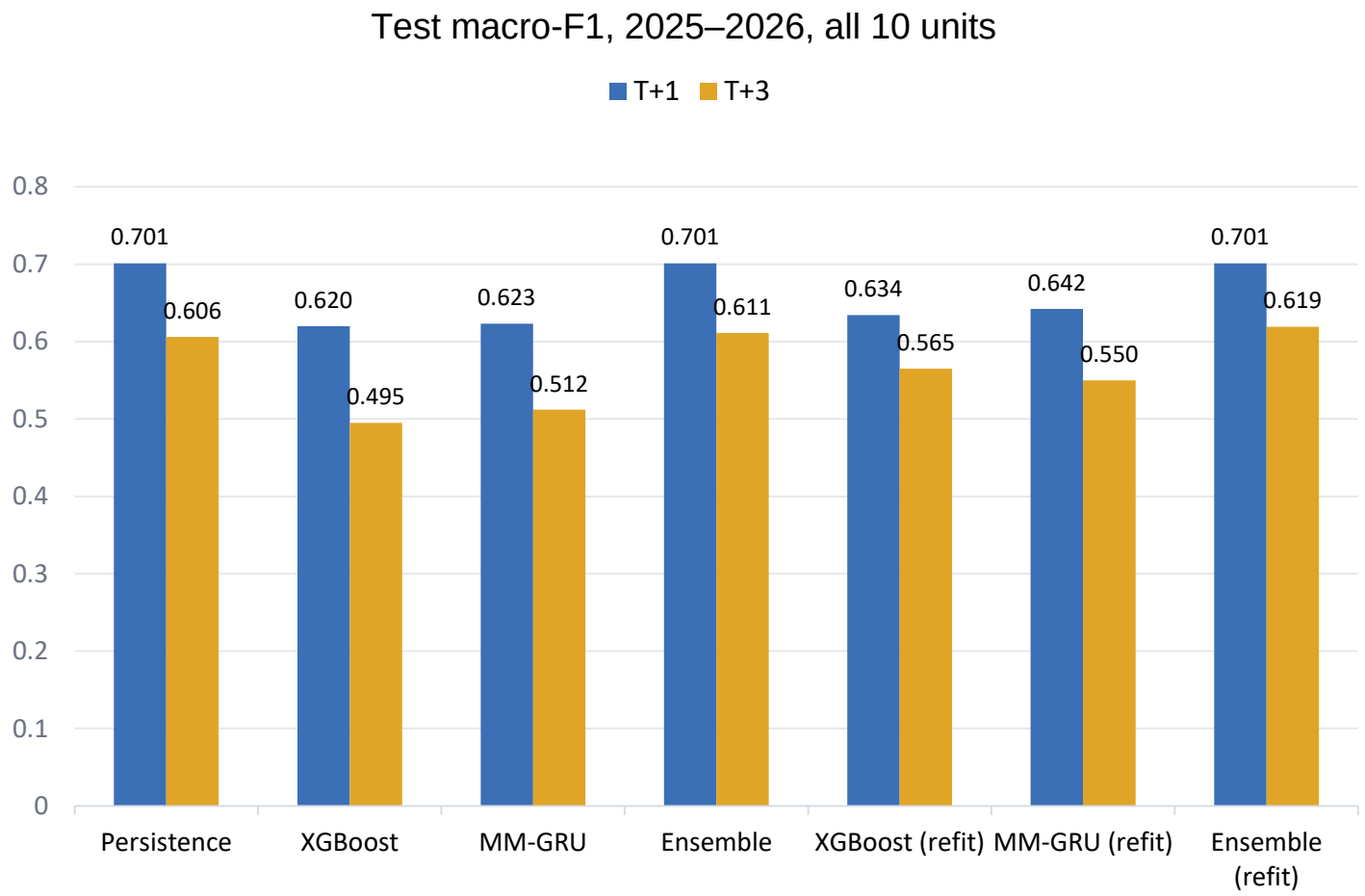
T+1: 0.123 → 0.040

T+3: 0.184 → 0.061

after temperature scaling the curve lies on the diagonal

RESULTS

Results: the ensemble ties persistence at T+1 and wins at T+3



0.701 vs 0.701

T+1 macro-F1: ensemble vs persistence (a tie)

0.619 vs 0.606

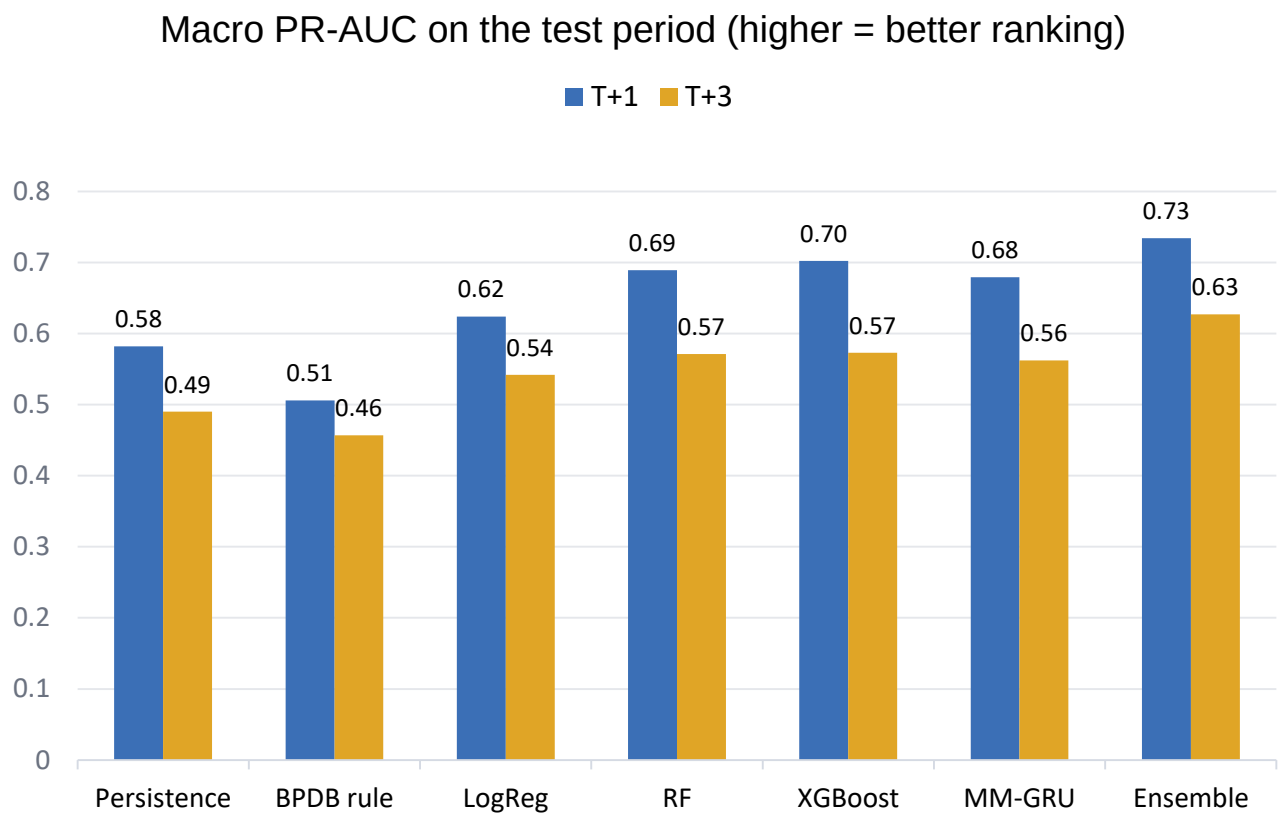
T+3 macro-F1: the ensemble beats persistence

0.77 vs 0.69

High-risk recall at T+1: the GRU catches more High days than persistence (but more false alarms)

RESULTS

Ranking and abstention: where the learned models really help



PR-AUC ignores the decision threshold and asks: are risky days ranked above safe days? Persistence cannot rank: LogReg, RF, XGBoost, the GRU and the ensemble all beat it.

With the abstain option (ensemble, refit)

62%

of unit-days answered at T+1 (the rest: 'Uncertain')

0.786

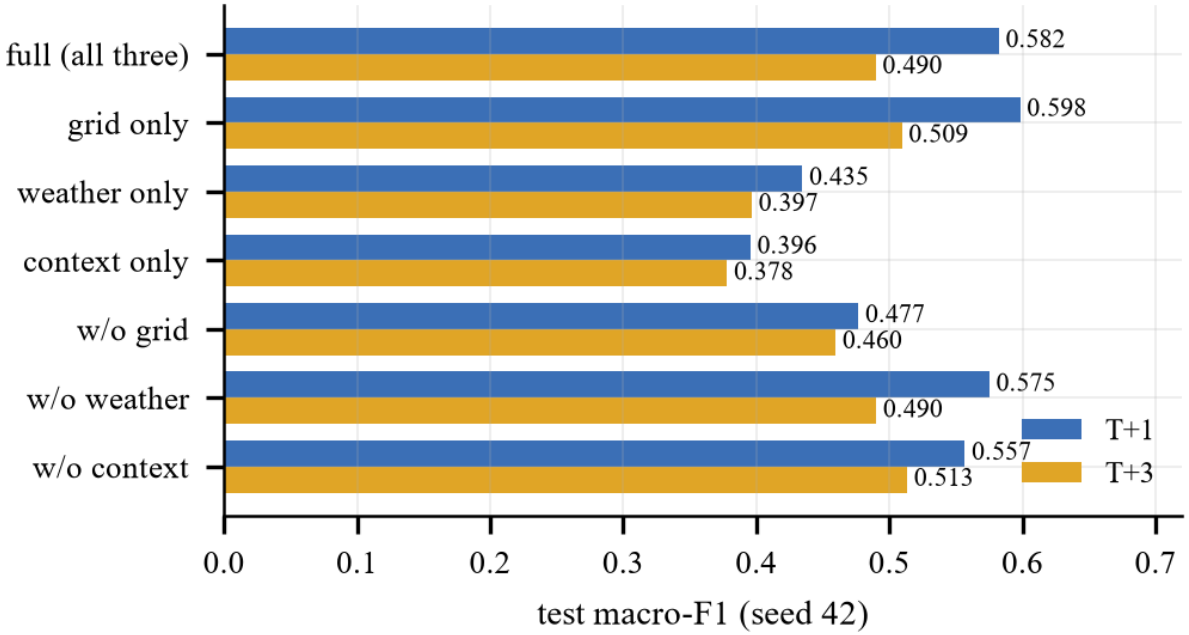
macro-F1 on answered days at T+1 (T+3: 0.720)

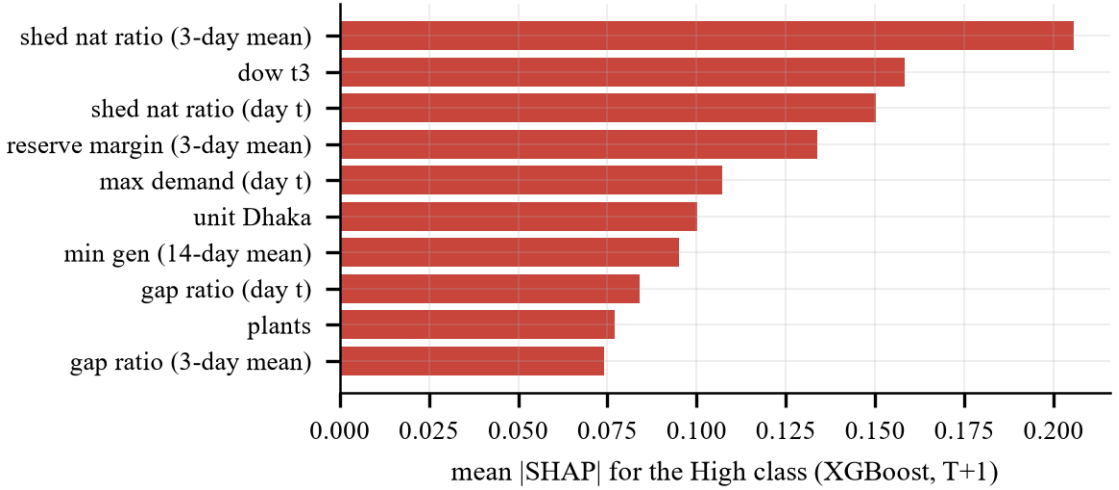
88%

of High-risk days caught at T+1 on answered days

RESULTS

What drives the risk? Grid data, then context and weather





Ablation

grid only 0.598 ≈ full 0.582; without grid 0.477

SHAP share (XGBoost, High)

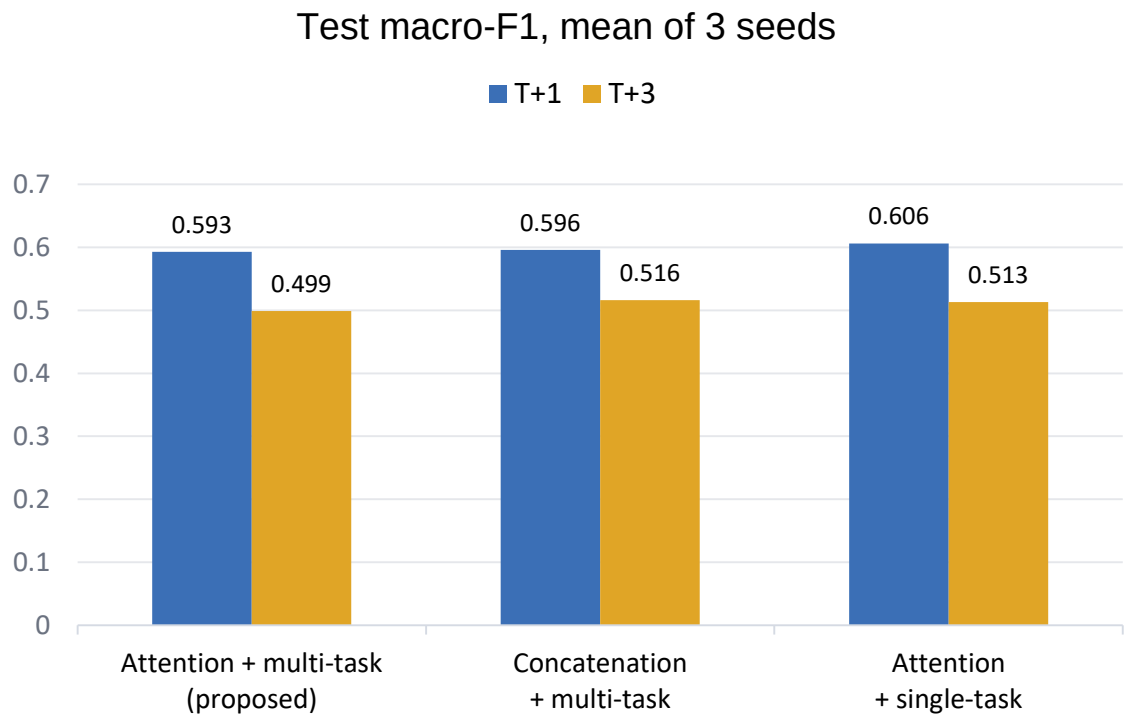
grid 65% · context 19% · weather 16%

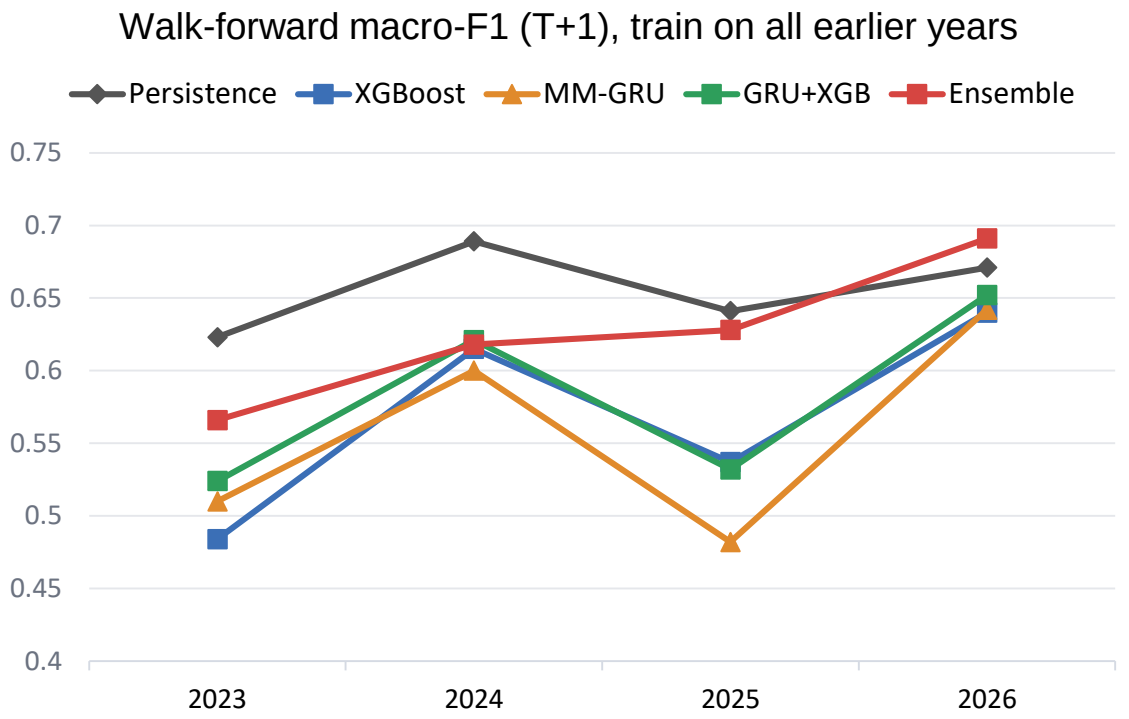
Attention ≠ importance

attention favours context (0.53) — it shows where the network looks, not what adds accuracy

LAB EXPERIMENTS

Lab experiments: fusion, multi-task and walk-forward

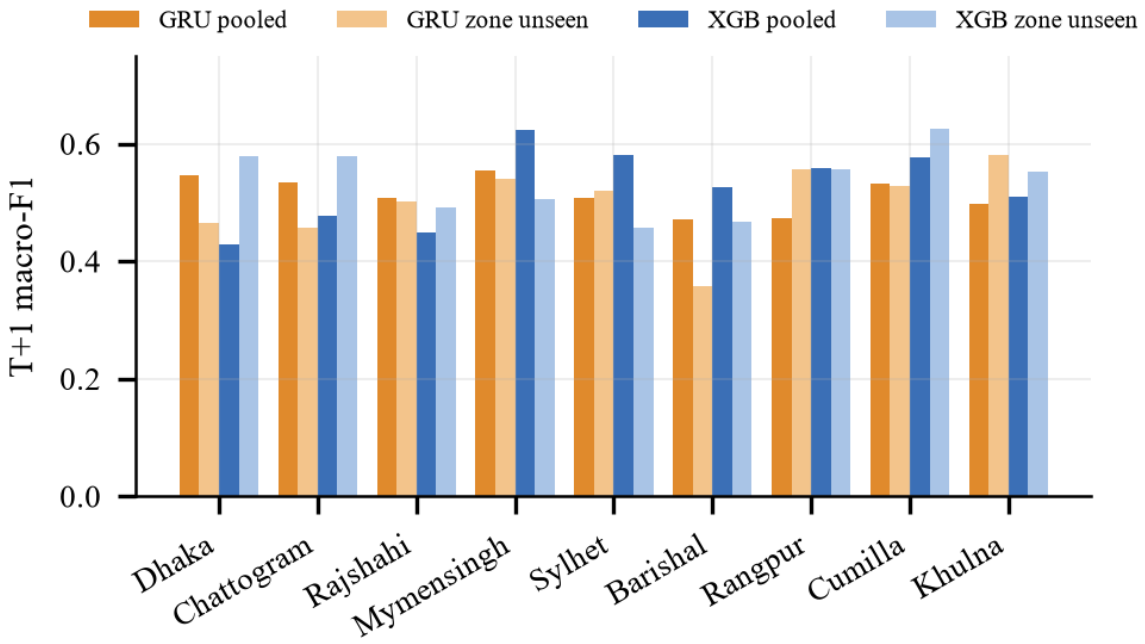




Fusion & task (T+1): attention 0.593 · concatenation 0.596 · single-task 0.606 — within seed noise. Walk-forward: persistence is best on average (0.656 vs ensemble 0.626), but the ensemble's gap closes every year and it leads in 2026 (0.691 vs 0.671); on PR-AUC the ensemble and the GRU beat persistence in every year.

RESULTS

Unseen zones and the live alert of 28 October 2026



Leave-one-division-out

A zone is removed from training and tested with an 'unknown zone' id. GRU mean macro-F1: 0.514 → 0.500 (T+1), 0.420 → 0.399 (T+3): it transfers to new zones.

Unit	T+1 (2026-10-29)	T+3 (2026-10-31)
National	High 0.63	High 0.52
Dhaka	Medium 0.73	Medium 0.64
Chattogram	Medium 0.56	Medium 0.49
Rajshahi	Medium 0.57	Uncertain (→Med)
Mymensingh	Uncertain (→Med)	Uncertain (→Med)
Sylhet	Medium 0.58	Medium 0.49
Barishal	Uncertain (→Med)	Uncertain (→Med)
Rangpur	Medium 0.57	Uncertain (→Med)
Cumilla	Medium 0.57	Medium 0.50
Khulna	Uncertain (→Med)	Uncertain (→Med)

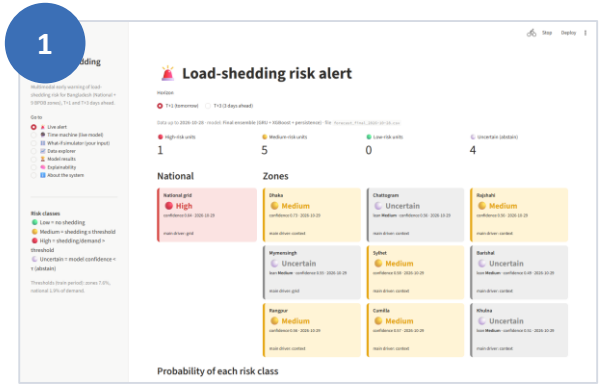
Deployed ensemble, refitted on all labelled data; number = confidence

DEMO

Live demo: Streamlit dashboard and Kaggle notebook

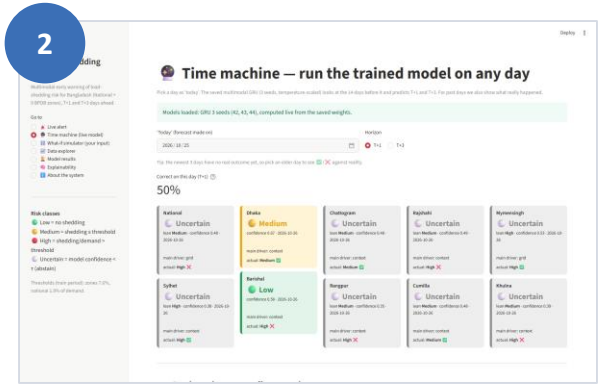
Streamlit dashboard (dashboard/app.py)

1



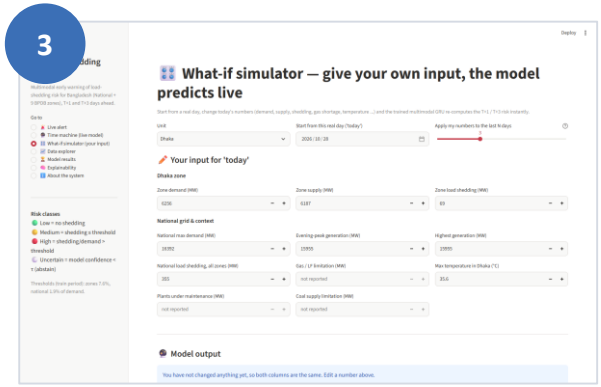
Live alert

2



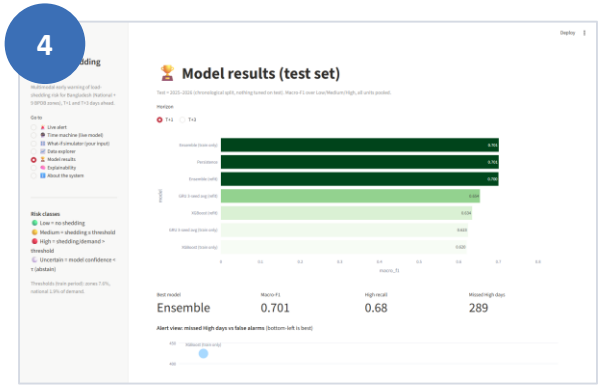
Time machine

3



What-if simulator

4



Results & explainability

Kaggle notebook

notebooks/kaggle_loadshed_pipeline.ipynb

- clones the GitHub repo, runs the leakage tests
- retrains baselines, GRU, ensemble, lab experiments
- shows every table and figure + a ‘time machine’ cell

Demo script (2–3 min)

1. Live alert → switch T+1 / T+3
2. Time machine → 15 Jul 2026, Mymensingh replay
3. What-if → Mymensingh shedding 300 MW for 3 days
4. Kaggle → show the finished run and outputs

CONCLUSION

A calibrated, explainable load-shedding risk warning for Bangladesh's grid and nine zones

Tie / win

vs persistence at T+1 / T+3

0.73 vs 0.58

macro PR-AUC: much better ranking

0.79 / 0.72

macro-F1 on the days it answers

0.040

ECE at T+1 after scaling

Future work

- same-day grid feeds instead of late PDF reports
- weather forecasts instead of reanalysis
- ordinal / cost-sensitive loss against severe misses
- regular walk-forward retraining

Limitations we report honestly

- persistence is very hard to beat on macro-F1
- strong distribution shift (2.4% → 27% High days)
- data-driven label threshold; late, imperfect reports
- attention/multi-task gave no accuracy gain

References

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Thank you

Questions and suggestions are welcome

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Code, data, dashboard and notebook: see repo.txt (public GitHub repository)